VIDHYA RANGARAJU, PhD

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Lab Webpage

CURRENT POSITIONS

Research Group Leader — From 01/2020
Max Planck Florida Institute for Neuroscience (MPFI), Florida, USA

Affiliate Faculty — From 10/2020
Charles E. Schmidt College of Science, Florida Atlantic University (FAU), Florida, USA

MATERNITY BREAKS

07/2020, 02/2023, 10/2025

EDUCATION AND TRAINING

Postdoctoral Fellow, Erin Schuman Lab — 02/2014 – 12/2019
Max Planck Institute for Brain Research (MPIBR), Germany

Ph.D. Tri-Institutional Ph.D. Program in Chemical Biology (TPCB), Timothy Ryan Lab — 07/2008 – 01/2014
Weill Cornell Medicine (WCM), New York
Cornell University, Ithaca, USA — 07/2007 – 06/2008

B.Tech. Bachelor of Technology, Industrial Biotechnology
Anna University, India — 07/2002 – 05/2006

GRANTS, FELLOWSHIPS, HONORS, AND AWARDS

McKnight Scholars Program Finalist	05/2025
Chan Zuckerberg Neuroscience Collaboration Supplement	12/2024 – 12/2025
NIH Director's New Innovator Award (DP2)	09/2024 – 08/2029
Society for Neuroscience Janett Rosenberg Trubatch Career Development Award	09/2024
CZI Ben Barres Early Career Acceleration Award	12/2023 – 11/2027
HHMI Freeman Hrabowski Scholars Program Finalist	03/2023
F.O.R.E Foundation Grant	07/2022– 12/2024
Louis D. Srybnik Foundation Grant	07/2021– 12/2024
Society for Neuroscience Peter and Patricia Gruber International Research Award	10/2020
Scientific discovery of the year, MPIBR	12/2019
GRADE Brain travel grant	09/2019
MacGillavry Fellowship, University of Amsterdam (declined)	04/2019
Lindau Nobel Laureate Meeting Award	06/2018
GlaxoSmithKline Stiftung travel grant	09/2015
EMBO Workshop travel grant	09/2015
Humboldt Research Fellowship for Postdoctoral Researchers (declined)	03/2015
Marie Curie Actions Individual Fellowship	01/2016 – 12/2017
EMBO Long-Term Postdoctoral Fellowship	01/2015 – 12/2015
Vincent du Vigneaud Award of Excellence, Vincent du Vigneaud symposium, WCM	04/2012
Best Poster Prize, Chemical Biology Student Symposium, Cornell University, USA	11/2007
Tri-Institutional Ph.D. Program in Chemical Biology Fellowship, WCM, USA	07/2007 – 05/2012
Junior Research Fellowship, National Centre for Biological Sciences, India	06/2006 – 06/2007
CSIR Program for Youth on Leadership in Science fellowship, CSIR, India	06/2001 – 06/2004
Chennai District Topper, State Board Examination, India	06/2000

ACTIVE FUNDING

NIH Director's New Innovator Award (DP2 MH140148)	$2,895,000	09/2024 – 08/2029
Carl Angus DeSantis Award	$ 262,500	07/2024 – 12/2027
Chan Zuckerberg Initiative Ben Barres Early Career Award	$1,200,000	12/2023 – 11/2027
Srybnik and F.O.R.E. Foundation Grant	$ 64,000	07/2021 – 12/2025
MPFI Annual General Operating Budget	$ 402,900/year	01/2020 – present

INVITED SCIENTIFIC TALKS AND CONFERENCES

University of Pennsylvania, Dept. of Physiology, Pennsylvania Muscle Institute, Philadelphia, USA 10/2026
Gordon Research Conference, Molecular and Cellular Neurobiology, Lucca, Italy 07/2026
Tata Institute of Fundamental Research, Sumitra Maharana lecture, Dept. of Biol. Sci., Mumbai, India 07/2026
Center for High Impact Neuroscience and Translational Applications (CHINTA), Kolkata, India 07/2026
UMDF Mitochondrial Medicine Conference, United Mitochondrial Disease Foundation, Orlando, USA 06/2026
Yale U, Wu Tsai Institute, Keynote Speaker, Grand Challenges - Energizing Cognition, New Haven, USA 06/2026
University of Colorado Anschutz, Department of Pharmacology, Aurora, USA 05/2026
University of Utah, School of Biological Sciences, Salt Lake City, USA 04/2026
Tulane University, Tulane Brain Institute Seminar Series, New Orleans, USA 04/2026
Lorentz Center Workshop, Fuel for Thought in Neurodevelopmental Disorders, Leiden, Netherlands 03/2026
Sunposium, MPFI's Biennial Neural Circuits Research Conference, West Palm Beach, USA 03/2026
Tata Institute of Fundamental Research, Society of Mitochondrial Research and Medicine, India 01/2026
Vanderbilt University-MPFI Symposium, Jupiter, USA 08/2025
Simons Collaborations on Brain Energetics Discussion Series, USA 05/2025
MITOTalks Seminar Series, USA and Europe 05/2025
Rutgers Robert Wood Johnson Medical School, Dept. of Neuroscience and Cell Bio., NJ, USA 03/2021, 05/2025
American Physiology Summit: Do we know how neurons work? Baltimore, USA 04/2025
Johns Hopkins University, Department of Cell Biology, Baltimore, USA 04/2025
Gordon Research Conference, Dendritic Mechanisms, Lucca, Italy, Ventura, USA 03/2017, 03/2025
Univ. of Texas Health McGovern Medical School, Dept. of Neurobiology and Anatomy, Houston, USA 01/2025
University of Miami Miller School of Medicine, Neuroscience Center Seminar Series, Miami, USA 12/2024
Max Planck Neuro Retreat, Frankfurt, Germany & Jupiter, USA 10/2022, 10/2024
Max Planck Biomedical Section Symposium, Harnack Haus, Berlin, Germany 10/2024
International Conference on Brain Energy Metabolism, Ljubljana, Slovenia 09/2024
Stony Brook University, Department of Neurobiology and Behavior, Stony Brook, USA 08/2024
Memorial Sloan Kettering Cancer Center, Mol. Pharm. & Chem. Bio. Seminar Series, New York, USA 08/2024
CZI Neuroscience 2024 Meeting, Monterey, USA 07/2024
EMBO Workshop, Principles of Dendritic Function and Computation, Chania, Greece 05/2024
CZI Neuroscience Program Kick Off, Redwood City, USA 03/2024
Advanced Neuroimaging Techniques, MPFI, Jupiter, USA 02/2024
CZI Neuroscience Early Career Acceleration Retreat, Redwood City, USA 02/2024
Friday Afternoon Reese Seminar Series, NIH, Maryland, USA 01/2024
Scientific Advisory Board Meeting, MPFI, Jupiter, USA 09/2021, 01/2024
CZI Ben Barres Early Career Acceleration Award Selection Symposium, Redwood City, USA 08/2023
Cytoskeleton of Neurons and Glia webinar, NIH, Maryland, USA 04/2023
HHMI Freeman Hrabowski Scholars Program Selection Symposium, Chevy Chase, USA 02/2023
Gordon Research Conference, Cell Biology of the Neuron, Waterville Valley, USA 06/2012, 06/2016, 06/2022
Indiana University, Gill Center for Biomolecular Science, Bloomington, USA 04/2022
Weill Cornell Medicine, Department of Psychiatry, New York, USA 05/2021
NeuroNex2, National Science Foundation Grant, Mitochondria Series Workshops, USA 02/2021
Johns Hopkins University, Department of Biological Chemistry, Baltimore, USA 12/2020
Dartmouth College, Department of Biological Sciences, Hanover, USA 11/2020
University of Colorado Boulder, Department of Biochemistry, Boulder, USA 11/2020
Johns Hopkins University, Watanabe Lab, Baltimore, USA 05/2020
Scripps Research, SYNAPSE 2020, Tri-Institutional Neuroscience Retreat, Jupiter, USA 01/2020
Society for Neuroscience, Chicago, Washington D.C., USA 10/2011, 11/2019
Bordeaux Neurocampus – Universite de Bordeaux, Bordeaux, France 07/2019
Institut Pasteur, Paris, France 06/2019
Max Planck Florida Institute for Neuroscience, Jupiter, USA 06/2019
Cell dynamics: Organelle-Cytoskeleton Interface, Lisbon, Portugal 05/2019
Universiteit van Amsterdam, Amsterdam, Netherlands 03/2019
University of Arizona, Department of Molecular and Cellular Biology, Tucson, USA 02/2019
Columbia University, Zuckerman Institute, New York, USA 02/2019
St. Jude Children's Research Hospital, Department of Developmental Neurobiology, USA 02/2019
Max Planck Research Group Leader Selection Symposium, Berlin, Germany 02/2019
The Francis Crick Institute, London, UK 02/2019
MRC Laboratory of Molecular Biology, Cambridge, UK 01/2019

University of Texas Southwestern Medical Center, Dept. of Neuroscience, Dallas, USA 12/2018
Max Planck Institute for Brain Research retreat, Frankfurt, Germany 09/2015, 09/2018
University of Lausanne, Dept. of Fundamental Neuroscience, Lausanne, Switzerland 06/2018
Young Investigators' Colloquium, Frankfurt, Germany 01/2018
EMBO Fellows' meeting, European Molecular Biology Laboratory, Heidelberg, Germany 06/2017
EMBO Workshop, Mitochondrial DNA and Neurodegeneration, Sitges, Spain 09/2015
EMBO Conference, RNA localization and local translation, Crete, Greece 06/2015
Daimler und Benz Stiftung, Ladenburg, Germany 01/2015
University of Cambridge, Dept. of Physiology, Development and Neuroscience, UK 10/2014
Genentech, Department of Neuroscience, South San Francisco, USA 08/2013
Max Planck Institute for Brain Research, Frankfurt, Germany 07/2013
Massachusetts Institute of Technology, Department of Chemistry, Boston, USA 12/2012
Keck Biomembrane retreat, Cornell University, Ithaca, USA 05/2012
Vesicle Trafficking Club, Weill Cornell Medicine, New York, USA 05/2013, 05/2011
Annual Vincent du Vigneaud Research Symposium, Weill Cornell Medicine, NY, USA 04/2012, 04/2011
Tri-Institutional PhD Program in Chemical Biology Retreat, Pennsylvania, USA 07/2011, 08/2010

GRANTS REVIEWED

Motor Neurone Disease Association, UK 09/2025, 08/2026
NIH Special Emphasis Panel, MMMND Study Section, Center for Scientific Review, USA 03/27/26 – 05/21/26
Human Frontier Science Program (HFSP) Postdoctoral Fellowships, France 10/2025
NIH, F03A Study Section, Center for Scientific Review, USA 01/2025 – 04/22/25
The Minerva Foundation GmbH, Germany 09/2024
NIH Early Career Reviewer, NDPR Study Section, Center for Scientific Review, USA 12/2023 – 02/29/24
Israeli Ministry of Innovation, Science, and Technology, Israel 08/2022, 09/2023
Israel Science Foundation, Isreal 01/2023
Human Frontier Science Program (HFSP) Research Grants, France 10/2022
Chan Zuckerberg Initiative Collaborative Pairs Program, USA 04/2022

PEER-REVIEWED SCIENTIFIC JOURNALS From 2010

Science, PNAS, Nature Biotechnology, Nature Neuroscience, Nature Communications, Nature Microbiology, Current Biology, Cell Reports, EMBO Reports, Current Opinion in Cell Biology, Cell Calcium, Molecular Neurobiology, iScience, Molecular and Cellular Biochemistry, Frontiers in Cell and Developmental Biology, PLOS Biology, Seminars in Cell and Developmental Biology, BMC Biology, Neurophotonics, Journal of Cellular Physiology, Biochemical Society transactions, Histology and Histopathology.

TEACHING EXPERIENCE

FAU Stiles-Nicholson Brain Institute, Neuroscience Seminar Series, USA 08/2026
Seventh Bangalore Cognition Workshop, Memory Module, Indian Institute of Science, India 06/2026
Advanced Neuroimaging Techniques, MPFI imaging course, USA 02/2024
Neuroscience Lecturer, IMPRS graduate program, MPIBR, Germany 11/2016 – 10/2017
Chemistry and Physics Tutor, International General Certificate of Secondary Education 04/2016 – 06/2017
Teaching Assistant, Max Planck Institute for Brain Research Teaching Lab, MPIBR, Germany 05/2014
Chemistry for Kids, WCM, USA 04/2013 – 05/2013

MENTORING EXPERIENCE (within brackets are current positions of alumni)

MPFI, USA From 01/2020
Postdoctoral Fellows – Nitheyaa Shree, Nimmy Varghese, Debalina Acharyya, Ilika Ghosh, Ruolin Fan, Sen Yang (Postdoc at Brown U)
Staff Scientist – Ruolin Fan
Ph.D. students – Ojasee Bapat, Monil Shah
Rotation students – Chelsea Mapp, Suhas Sunke (PhD at MPFI), Aidan McDonald (PhD at FAU)
Master's student – Rasya Bhat (Res. Asst. at CHINTA)
Postbaccalaureates – Riya Sivakumar (PhD at Harvard), Danyu Luo (PhD at Northwestern), Ojasee Bapat (PhD at MPFI), Jacob LaMar (MS at FAU)
Research technicians – Juan Patarroyo, Devahuti Bhuyan (PhD at U Alabama), Sabrina Perez (Northrop Grumman), Shaun Foutch (MS at MPFI)
Undergraduates – Jean-Manuel Mena (BS at Cornell), Tejas Purimetla (MD at Dartmouth), Nicholas Moy

(BS at FAU), Neha Anand (BE at IIT), Mohan Peddada (BS at Johns Hopkins), Kevin Pena
High school students – Luca Pishos (BE at U Florida), Gayatri Sathyanarayan (BE at Cornell), Jake Zur (BE at U Michigan)
Visiting Scientists – Sen Yang (Postdoc at Brown U), Jenna Wingfield (Postdoc at UF Scripps), Ken Kunugitani (MD PhD at Kyoto University)

Schuman lab, Department of Synaptic plasticity, MPIBR, Germany 02/2015 – 12/2019
Master's students – Sarah Kruessel (Postdoc at U Heidelberg), Ana Alupei (Fintiba GmbH)
Research Technician – Christina Thum (Tech. at MPIBR)
High school students – Anna Sawyer, Rauf Gunsay, Johanna Möller (MS at Heidelberg U), Jessica Patzelt, Shreya Srinivasa, Rebeca Tenney

Ryan lab, Department of Biochemistry, WCM, USA 01/2011 – 12/2013
Postdoctoral Fellows – PingYue Pan (PI at Rutgers U), Jaime de Juan-Sanz (PI at Paris Brain Institute)
Ph.D. student – Kate Gao (Pi Health)

MENTEE AWARDS AND HONORS

Invited Speaker, MSN seminars, Mount Sinai – Ilika Ghosh 01/2027
Invited Speaker, P-SPINE Fellow, University of Pennsylvania – Ilika Ghosh 10/2026
Selected Participant, CZI Faculty Application Bootcamp – Ilika Ghosh 06/2026
Best Talk Award and Selected Speaker, The NeuroCollective Symposium – Monil Shah 05/2026
Minisymposia Speaker, Travel Grant Awardee, Elevator Pitch Awardee, ASCB 2025 – Ilika Ghosh 12/2025
Poster Presenter, CZI Imaging the Future – Monil Shah 09/2025
Invited Speaker, ThermoFisher Beyond the Beam Seminar Ser. on Electron Tomography – Monil Shah 10/2024
Selected Speaker, Japan-MPFI Multi-D Analysis of Memory Mechanisms Retreat – Monil Shah 09/2024
Janelia Visiting Scientists, Lippincott-Schwarz Lab – Ojasee Bapat and Monil Shah 09/2024
Carl Angus DeSantis Award – Ilika Ghosh 07/2024 – 12/2027
Selected for EMBO Intensive Course in Electron Microscopy for Cell Biology – Monil Shah 06/2024
Invited Speaker, CHINTA, TCG Crest – Ilika Ghosh 05/2024
Teaching Assistants, MPFI imaging course – Ilika Ghosh, Monil Shah 02/2024
Best Poster Award, MPFI Retreat – Monil Shah 10/2023
Invited Speaker & Best Oral Presentation Award, UF Scripps Research Fest – Ilika Ghosh 05/2023
Travel Grant, NeuroNex Workshop – Monil Shah 05/2023
Best Poster Award, Zeiss Imaging Applications for Neuroscience Summit – Monil Shah 05/2023
Invited Speaker & Best Question Award, Florida Consortium on the Neurobio. of Cognition – Ruolin Fan 05/2023
Outstanding OWL Award and Outstanding Thesis Award – Tejas Purimetla 05/2023
Society for Neuroscience Trainee Professional Development Award – Ilika Ghosh 11/2022
Society for Neuroscience Invited Nanosymposium Speaker – Ilika Ghosh 11/2022
Nambu Memorial Summer Research Experience Program – Tejas Purimetla 05/2022 – 08/2022

THESIS COMMITTEE MEMBER/EXAMINER

Ceylin Zeybek, Yasuda Lab, MPFI, USA From 05/2026
Valentina Villani, Sun Lab, DANDRITE, Aarhus University, Denmark From 03/2026
Michael Maibach, Lohmann Lab, Netherlands Institute for Neuroscience, Netherlands From 01/2025
Brittni R Walker, Moraes Lab, University of Miami Miller School of Medicine, USA 07/2024 – 08/2024
Yiwei Hua, Tian Lab, MPFI, USA From 04/2024
Carlos C. Ramos, Morais Lab, Universidade de Lisboa, Portugal 01/2024 – 04/2024
Yibo Zhao (cosponsor on her F31), Puthanveettil Lab, UF Scripps, USA From 07/2023

RECRUITMENT

Okinawa Science and Technology, Faculty Candidate Review, Okinawa, Japan 02/2025 – 03/2025
UF Scripps Neuroscience Faculty Recruitment Committee, Florida, USA 06/2022 – 2023
International Max Planck Research School (IMPRS) – MPFI graduate program From 02/2020
International Max Planck Research School (IMPRS) – MPIBR graduate program 02/2018
Princeton Graduate and Professional School Fair, Princeton, USA 10/2011
Annual Biomedical Research Conference for Minority Students, North Carolina, USA 11/2010

INVITED PANELS AND WORKSHOPS

Alumni Speaker and Panelist, TPCB Retreat & TPCB25 Alumni Day, USA 09/2023, 09/2026

Simons Collaboration on Brain Energetics and Metabolic Constraints, USA 05/2026
Session Chair, Novel Measurements of Neural Circuit Function, Sunposium, USA 03/2026
iJOBS career seminar speaker, Rutgers Robert Wood Johnson Medical School, USA 05/2025
Celebration of Women in Neuroscience, Society for Neuroscience, USA 10/2024
Session Chair, Organelle Biology and Metabolism, CZI Neuroscience 2024 Meeting, USA 07/2024
Panel Member, Future of Dendritic Research, EMBO Workshop, Greece 05/2024
Work-Life Balance Panel Member, Network for Women in Science, MPFI, USA 04/2022
Neuroscience Career Skills Panel Member, Johns Hopkins School of Medicine, USA 02/2022
Research Career Panel Member, MPFI Postbaccalaureate Program, USA 02/2021
External Advisory Committee Member, NeuroNex2, National Science Foundation grant, USA From 01/2021

ORGANIZATION OF MEETINGS

Max Planck NeuroMEETS – Neuroscience Meetings for Extramural Emerging Talents, MPFI, USA From 06/2022
Max Planck Career Day, MPIBR, Germany 06/2019
Trafficking Club, Department of Biochemistry, WCM, USA 08/2011 – 05/2012

SERVICE

External Advisory Editorial Board Member, Current Biology, USA From 11/2026
NIH F and K Grants Peer Review Member, MPFI, USA From 05/2026
Poster Judge, The NeuroCollective (MPFI, UF Scripps, FAU) Symposium 05/2026
External Advisory Editorial Board Member, EMBO Reports, Germany From 06/2025
IACUC Scientific Member, MPFI, USA 11/2023 – 01/2025
Selection Committee Member, Peter Gruss Young Investigator Award, MPFI, USA From 05/2022
Founder and Selection Committee Member, Max Planck NeuroMEETS, MPFI, USA From 06/2022
Founder and Postdoctoral Association Representative, MPIBR, Germany 10/2017 – 06/2019
Academic team member, DNO, German Neuroscience Competition, Germany 09/2016 – 04/2018
Vice President, Weill Cornell Consulting Club, Weill Cornell Medical College, USA 01/2012 – 07/2012
Student Buddy, Expanding Your Horizons, Cornell University, Ithaca, USA 04/2008

SCIENCE OUTREACH

Brain Ponderings Podcast with Mark Mattson, USA 05/2026
South Florida PBS Town Hall, Memory and the Mind, USA 03/2026
South Florida PBS (Public Broadcasting Service) Feature on 'The Resilient Brain', USA 12/2025
FAU Media Relations, Rangaraju Lab IMPRS Graduate Students, MPFI, USA 07/2025
Music and Mind with Opera Legend Renée Fleming, Panel Member, Kravis Center, USA 01/2025
Science Meets Music, Public Speaker, The Benjamin School, USA 02/2024
Board of Trustees Meeting Public speaker, MPFI, USA 05/2023
MPFI Institute Breakfast Public Speaker, USA 12/2022
Science Career Day, Rangaraju Lab, MPFI, USA 09/2022
Brain Exploration Day, Rangaraju Lab, MPFI, USA 08/2022
Brain Bee Winners – Day at the Rangaraju Lab, MPFI, USA 06/2022
Board of Trustees Meeting Public speaker, MPIBR, Germany 12/2019
Neuroscience Study Guide Author/Editor, DNO-German Neuroscience Competition 09/2016 – 04/2018
Night of Science Public speaker, Goethe University, Germany 06/2018
'Bar of Science' Public speaker, MPIBR, Germany 05/2017
Neuroscience Graduates' Colloquium speaker, Mainz, Germany 05/2017
Ivy Circle and Cornell Alumni Club, Host and Public Speaker, Frankfurt, Germany 04/2015, 04/2016
Cornell Student Outreach, WCM, USA 03/2012

ADDITIONAL TRAINING

Institutional Animal Care and Use Committee (IACUC) online training, MPFI, USA 02/2020
EMBO Training Course on Negotiation for Scientists, Germany 10/2019
Optical Imaging and Electrophysiological Recording in Neuroscience, France 05/2018
EMBO Laboratory Management course for Postdoctoral Fellows, Germany 10/2016
Targeted Drug Discovery Workshop, National Hellenic Research Foundation, Greece 10/2016
FELASA course on Laboratory Animals, Germany 06/2016 – 07/2016

OTHER POSITIONS

Project Assistant, Baird Lab, Cornell University, USA	01/2008 – 05/2008
Junior Research Fellow, National Centre for Biological Sciences, India	06/2006 – 05/2007
Research Assistant, National Centre for Biological Sciences, India	05/2005 – 04/2006
CPYLS Fellow, Central Leather Research Institute, India	05/2004 – 06/2004

IN THE MEDIA

Brain Ponderings Podcast with Mark Mattson (YouTube, Spotify)
South Florida PBS Documentary, The Resilient Brain (link)
South Florida PBS Town Hall, Memory and the Mind (link1, link2)
PreLights: Synapses Drive Local Mitochondrial ATP Synthesis to Fuel Plasticity (link)
SfN 2024 Celebration of Women in Neuroscience (YouTube)
Stories of WiN, Podcast and Interview (link)
How are Memories Powered? Public Lecture at 'Science Meets Music' (YouTube)
How Mitochondria Stay Still in Neurons, article in 'The Scientist' (link).
Molecular Adaptations of Brain Energy (YouTube)
ALS-linked Protein VAP Stabilizes Mitochondria to Support Memory Formation (YouTube)
A Molecular Anchor: Scientists Identify ALS-linked Protein VAP as Key Stabilizer of Mitochondria (link)
PreLights: VAP Spatially Stabilizes Dendritic Mitochondria to Locally Fuel Synaptic Plasticity (link)
Celebrating Women Making History at MPFI (link)
Water Cooler Neuroscience Podcast, How Neurons Use Energy with Dr. Vidhya Rangaraju (Apple)
MPFI News, Meet the Scientist: Vidhya Rangaraju (link)

PATENT CONTRIBUTION

Quantitative genetically encoded optical reporter construct. U.S. 9487819 B2; Addgene 51819 (>50 requests)

PREPRINTS (lab members; #high school students, undergraduates, post-baccalaureates, master's students)

2. Mitochondria remodel near synapses to boost metabolic capacity and fuel plasticity
Monil Shah, Ilika Ghosh, Nitheyaa Shree Ramesh, Luca Pishos#, Tristano Pancani, Valentina Villani, Ryohei Yasuda, Sahil Loomba, Chao Sun, Naomi Kamasawa, **Vidhya Rangaraju***
bioRxiv doi: 10.1101/2025.08.27.672715v3 (2026), *under review*. Number of citations: 2. *Corresponding author. PMID: 40909516. PMCID: PMC12407993

1. An expanded palette of bright and photostable organellar Ca2+ sensors
Agathe Moret, Helen Farrants, Ruolin Fan, Kelsey G. Zingg, Bryon Silva, Camilla Roselli, Thomas G. Oertner, Christine E. Gee, Dafni Hadjieconomou, **Vidhya Rangaraju***, Eric R. Schreiter, Jaime de Juan-Sanz
bioRxiv doi: 10.1101/2025.01.10.632364 (2025). Number of citations: 5.
eLife14:RP107845 doi: 10.7554/eLife.107845.1 (2025), *in revision.* *Senior author.

PUBLICATIONS (NIH My Bibliography) (lab members; #undergraduates, post-baccalaureates, master's students)

13. Synaptic inputs drive local mitochondrial ATP synthesis to fuel plasticity
Ilika Ghosh, Ruolin Fan, Monil Shah, Ojasee Bapat, **Vidhya Rangaraju***
bioRxiv doi: 10.1101/2025.04.09.648032 (2025). Number of citations: 12.
Prelights doi: 10.1242/prelights.40383.
Nature Neuroscience*, in press.* *Corresponding author

12. Activity-regulated circSamm50 modulates mitochondrial dynamics and spine structural plasticity
Kaushik Chanda, Ojasee Bapat, Jenna Lynne Wingfield, Yosef Avchalumov, Melissa Kazantzis, Jackson P Carter, Neelam Sharma, Ronald Davis, Jason X-J Yuan, **Vidhya Rangaraju***, Sathyanarayanan V Puthanveettil
SSRN [Preprint] doi: 10.2139/ssrn.5396301 (2025).
Cell Reports 45, 117378 (2026). *Senior author. PMID: 42228570.

11. Mitochondrial Ca2+ efflux controls neuronal metabolism and long-term memory across species
Anjali Amrapali Vishwanath, Typhaine Comyn, Chaitanya Chintaluri, Carla Ramon-Duaso, Ruolin Fan, Riya Sivakumar#, Mario Lopez-Manzaneda, Thomas Preat, Tim P. Vogels, **Vidhya Rangaraju***, Arnau Busquets-Garcia, Pierre-Yves Placais, Alice Pavlowsky, Jaime de Juan-Sanz
bioRxiv doi: 10.1101/2024.02.01.578153 (2024).
Nature Metabolism (2026) doi: 10.1038/s42255-026-01451-w. Number of citations: 28. *Senior author.

10. Periodic ER-plasma membrane junctions support long-range Ca2+ signal integration in dendrites
Lorena Benedetti, Ruolin Fan, Aubrey V. Weigel, Andrew S. Moore, Patrick R. Houlihan, Mark Kittisopikul, Grace Park, Alyson Petruncio, Philip M. Hubbard, Song Pang, C. Shan Xu, Harald F. Hess, Stephan Saalfeld, **Vidhya Rangaraju***, David E. Clapham, Pietro De Camilli, Timothy A. Ryan, Jennifer Lippincott-Schwartz
bioRxiv doi: 10.1101/2024.05.27.596121 (2024).
Cell 188, 484 (2025). Number of citations: 70, *Senior author. PMID: 39708809.

9. Synaptically-targeted long non-coding RNA SLAMR promotes structural plasticity by increasing translation and CaMKII activity
Isabel Espadas, Jenna L. Wingfield, Yoshihisa Nakahata, Kaushik Chanda, Eddie Grinman, Ilika Ghosh, Karl E. Bauer, Bindu Raveendra, Michael A. Kiebler, Ryohei Yasuda, **Vidhya Rangaraju***, Sathyanarayanan Puthanveettil
Res. Sq. [Preprint] doi: 10.21203/rs.3.rs-2489387/v1 (2023). Number of citations: 3
Nature Communications 15, 2694 (2024). Number of citations: 31, *Senior author. PMID: 38538603. PMCID: PMC10973417.

8. NMNAT2 supports vesicular glycolysis via NAD homeostasis to fuel fast axonal transport
Sen Yang, Zhen-Xian Niou, Andrea Enriquez, Jacob LaMar#, Jui-Yen Huang, Karen Ling, Paymaan Jafar-Nejad, Jonathan Gilley, Michael P Coleman, Jason M Tennessen, **Vidhya Rangaraju***, Hui-Chen Lu
Res. Sq. [Preprint] doi: 10.21203/rs.3.rs-2859584/v1 (2023). Number of citations: 3
Molecular Neurodegeneration 19, 13 (2024). Number of citations: 38, *Senior author. PMID: 38282024. PMCID: PMC10823734.

7. VAP spatially stabilizes dendritic mitochondria to locally support synaptic plasticity
Ojasee Bapat, Tejas Purimetla#, Sarah Kruessel#, Monil Shah, Ruolin Fan, Christina Thum, Fiona Rupprecht, Julian D. Langer, **Vidhya Rangaraju***
bioRxiv doi: 10.1101/2023.01.16.524245 (2023). Number of citations: 4
Prelights doi: 10.1242/prelights.33633.
Nature Communications 15, 205 (2024). Number of citations: 58, *Corresponding author. PMID: 38177103. PMCID: PMC10766606.

6. The Prevalence and Specificity of Local Protein Synthesis during Neuronal Synaptic Plasticity
Chao Sun, Andreas Nold, Claudia M Fusco, **Vidhya Rangaraju***, Tatjana Tchumatchenko, Mike Heilemann, Erin M. Schuman
bioRxiv doi: 10.1101/2020.04.29.068833 (2020). Number of citations: 4
Science Advances 7, eabj0790 (2021). Number of citations: 133**,** *Senior author. PMID: 34533986. PMCID: PMC8448450.

5. Pleiotropic mitochondria: The influence of mitochondria on neuronal development and disease
Vidhya Rangaraju, Tommy L. Lewis Jr, Yusuke Hirabayashi, Matteo Bergami, Elisa Motori, Romain Cartoni, Seok-Kyu Kwon, Julien Courchet
Journal of Neuroscience 39, 8200 (2019). Number of citations: 325. PMID: 31619488. PMCID: PMC6794931.

4. A genetically encodable cell-type-specific protein synthesis inhibitor
Maximilian Heumüller, Caspar Glock, **Vidhya Rangaraju**, Anne Biever, Erin M Schuman
Nature Methods 16, 699 (2019). Number of citations: 21. PMID: 31308551.

3. Spatially stable mitochondrial compartments fuel local translation during plasticity
Vidhya Rangaraju, Marcel Lauterbach, Erin M. Schuman
Cell 176, 73 (2019). Number of citations: 465. PMID: 30612742.
Special commentary in *Neuron* 101, 204 (2019)

2. Local translation in neuronal compartments: how local is local?
Vidhya Rangaraju*, Susanne tom Dieck, Erin M. Schuman
EMBO reports 18, 693 (2017), Invited review, Number of citations: 243, *Corresponding author. PMID: 28404606. PMCID: PMC5412868.

1. Activity-driven local ATP synthesis is required for synaptic function
   Vidhya Rangaraju, Nathaniel Calloway, Timothy A. Ryan
   Cell 156, 825 (2014). Number of citations: 1039. PMID: 24529383. PMCID: PMC3955179.